DEMO APP · CONCEPT V0.1

Baghbaan

Everything a Kashmiri apple grower needs, in one app: find what is wrong, buy the right medicine, learn, sell well and book help.

- Orchard Doctor: Yes / No diagnosis
- Genuine inputs
- Mandi, highway & weather
- Trucks & labour
- Kashmiri · Urdu · Hindi · English
- Works offline



Demo app walkthrough

November 2026 · All names, prices, rates and headlines are sample data.
Clickable demo: claude.ai/artifact/T7wV1p9vo63Sqa659bEQCG

THE IDEA

One app for the whole orchard year

Today a grower asks the dealer what to spray, hears prices from the commission agent, follows the highway on WhatsApp and phones around for trucks and labour. Baghbaan puts all of it in one place, built around the first question every grower has: what is wrong with my trees, and what do I do?

Orchard Doctor

The main page. Yes / No and picture questions find the cause of a problem, then say what to do now and how to stop it next year. Photo check and a human expert back it up.

Buy

Sprays, fertilizers, organic inputs, tools, nets, packing and saplings from licensed dealers. Dose calculator, group rotation, genuine-pack check and pay-after-harvest.

Learn

Three-minute lessons with Kashmiri audio: soil, planting, high-density orchards, pruning, pests, safe spraying, harvest, selling and schemes. Saved for offline use.

Mandi & news

Daily mandi rates, a where-to-sell calculator, NH-44 and Mughal Road status, rail parcel, spray-window weather, disease risk and advisories.

Services

Book trucks and shared loads, labour teams, spray services, tractors, expert visits, soil tests, cold storage, beehives and buyers. Verified and rated.

My orchard

Blocks, varieties, spray diary, costs and harvest records. It fills itself from what the grower already does, and the grower controls who sees it.

HOW THE PARTS WORK TOGETHER

Spot a problem
Home alert or own eyes

›

Doctor finds it
2–5 Yes / No questions

›

Buy the right spray
Rotated group, exact dose

›

Book a spray team
Verified, rated

›

Saved to diary
Reused next time

›

Sell at best mandi
Net price after freight

WHO IT IS FOR

Built for the grower with a basic phone and little time

Three illustrative growers shaped the design. None of them wants to fill a form. All of them want an answer they can trust, in their own language.

Ghulam Nabi, 58

GN

Wachi, Shopian · 25 kanal, traditional + new high-density block

SPEAKS	Kashmiri; reads Urdu slowly. His son set up the phone.
WORRIES	Scab after rain, fake medicine, the highway closing at harvest.
USES	Voice, Yes / No Doctor, spray alerts, truck booking.

Insha, 27

IN

Sopore · 12 kanal high-density, family orchard

SPEAKS	Urdu, English; agriculture graduate.
WORRIES	Exact schedules for M9 plants, mandi spreads, records for buyers.
USES	Spray diary, where-to-sell, expert chat, lessons.

Abdul Rashid, 45

AR

Kulgam · 6 kanal, sells to a pre-harvest contractor

SPEAKS	Kashmiri; reads very little. Prefers phone calls.
WORRIES	Credit from the dealer, labour at harvest, fair prices.
USES	Voice helpline, pay after harvest, labour teams.

SIX DESIGN RULES

Voice and pictures first

Every question has a picture and a Listen button. In studies, low-literacy users finished no tasks with text menus but most with voice or pictures.

Works when the network does not

J&K has seen hundreds of internet shutdowns. Lessons, schedules and saved rates work offline; SMS and a voice line are backups.

Trust is built in

Licensed dealers only, genuine-pack checks, verified providers, sources shown on every rate and advisory.

Ask little, give back first

No signup form. One question at a time, only when it improves the answer, always with a reason and a Skip.

Season-aware

The home screen follows the crop stage at the grower's altitude: dormant oil in March, scab in spring, mites in summer, urea at leaf fall.

Local words and units

Kanal, peti (box), 200 L drum, ₹ per kg and per box, district and village names the grower uses.

APP MAP

Five tabs, one main question

The bottom bar has five tabs. Home opens on the Orchard Doctor; every other tab links back to it, and its answers link out to products, lessons and services.

Home · Orchard Doctor

- Today: alerts, crop stage, rates, highway
- Start: where is the problem?
- Yes / No and picture questions
- Result: cause, do now, prevent
- Photo check (beta)
- Ask an expert · voice note
- One quick question a day

Buy

- Kit made for your orchard
- Shop by need (12 categories)
- Filter by problem and stage
- Product: brands, dose, safety
- Order: pickup or delivery
- Pay: UPI, cash, after harvest
- Genuine-pack QR check

Learn

- Continue a course
- For this season
- 12 topics, 120+ lessons
- Soil sampling lesson + quiz
- Pests & diseases library
- Orchard year calendar
- Saved for offline

Mandi & news

- Rates by variety and grade
- 30-day trend
- Where should I sell?
- Highway, Mughal Road, rail
- Weather and spray window
- Disease risk, hail alerts
- Advisories and schemes

Services

- Trucks and shared loads
- Labour and spray teams
- Tractor and equipment
- Expert visit, soil test
- Cold storage (CA)
- Beehives, nursery plants
- Track a booking

My orchard

- Blocks & varieties
- Spray diary
- Costs & income
- Harvest record
- Pest map near me
- Privacy & data

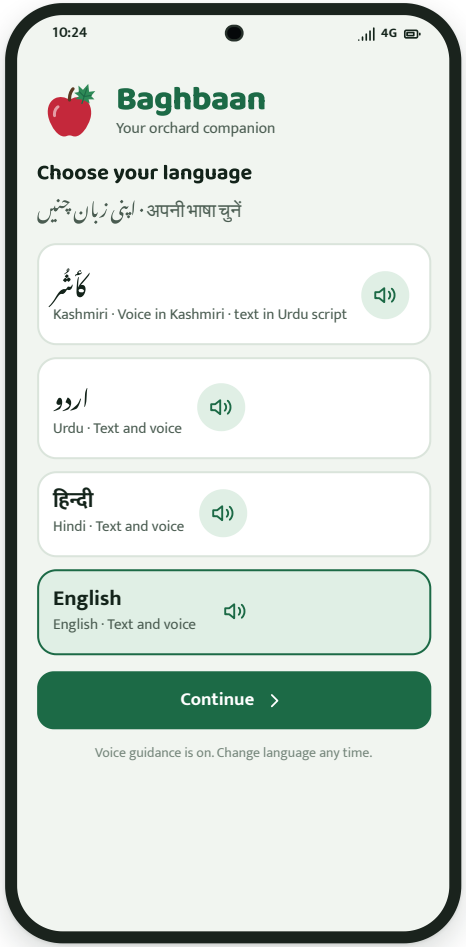
Other ways in

- WhatsApp / SMS keywords
- Missed-call voice line in Kashmiri
- Assisted at Kisan Khidmat Ghars
- Village helpers (Bagh Mitras)
- Expert desk for advisors

GETTING STARTED

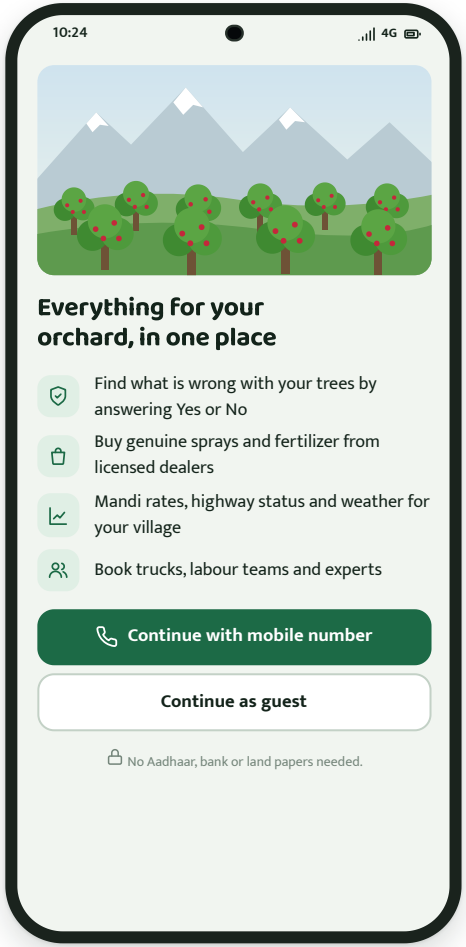
Ready in three taps, no form to fill

The grower picks a language, skips signup if they like, and names a village. That is all the app needs to give local advice.



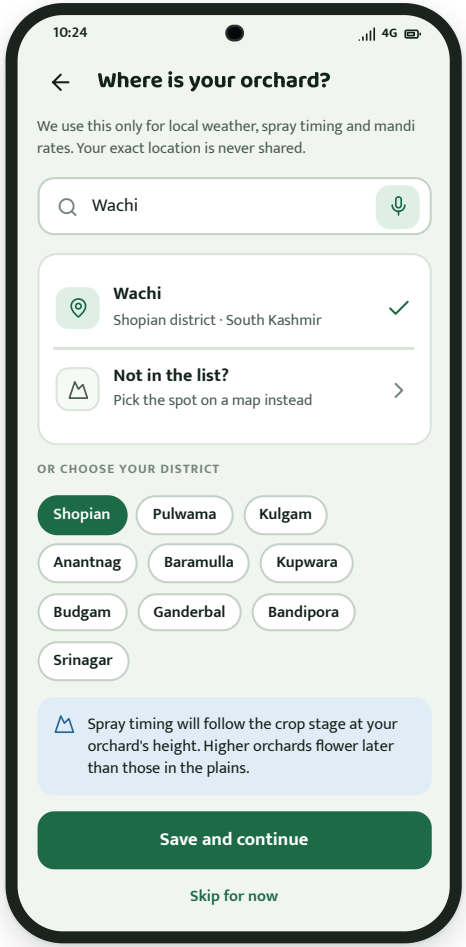
1 Choose a language

Kashmiri means Kashmiri voice with Urdu-script text, because more growers read Urdu than written Kashmiri. Every option can be heard.



2 Guest mode is fine

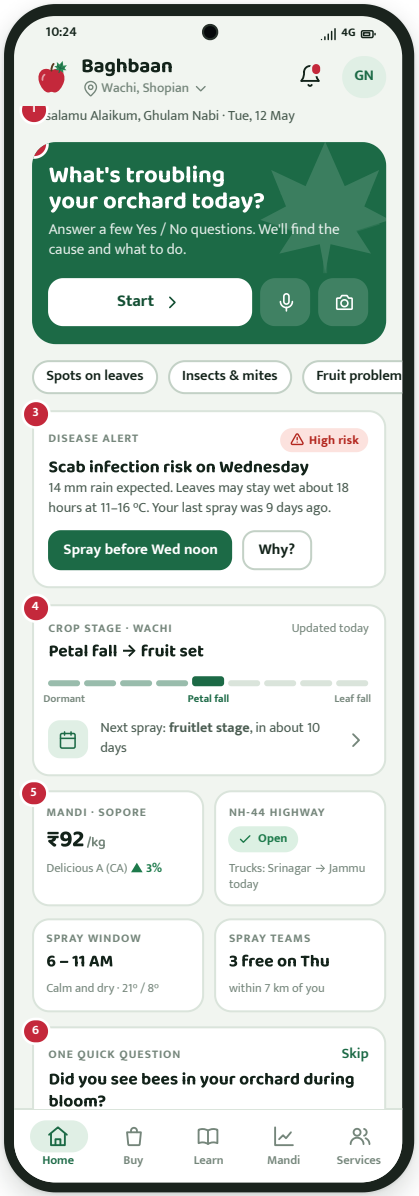
Mobile number is optional. No Aadhaar, bank or land papers are ever asked for.



3 Village, not GPS

The village sets weather, crop-stage timing and mandi rates. Exact location stays on the phone.

The home screen answers "what should I do today?"



1. **Greeting in the grower's language**
   Name only if they chose to give it. Date and village keep every message local.

2. **The main question, front and centre**
   Start opens the Orchard Doctor. The microphone takes a spoken problem in Kashmiri; the camera starts a photo check.

3. **Alerts that come with an action**
   Rain plus leaf wetness plus temperature gives a scab infection risk for the grower's own area. The button goes straight to the right sprays, rotated away from the last group used.

4. **Crop stage for this altitude**
   Higher orchards flower later. The strip shows where the season is and what spray is due next, read from the spray diary.

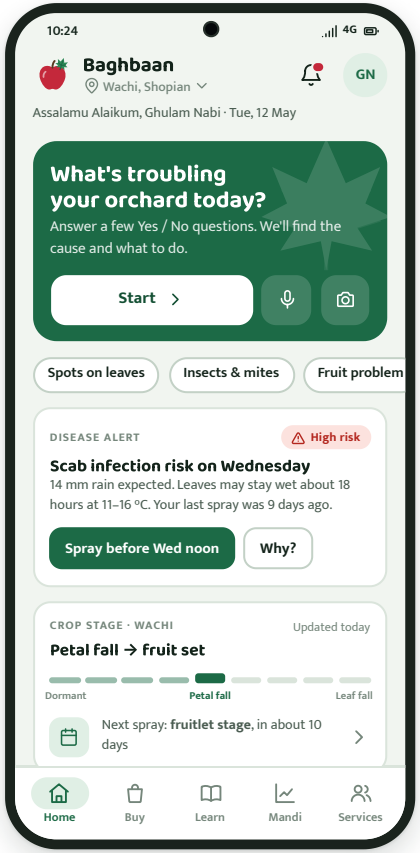
5. **Today's numbers**
   Mandi rate with trend, NH-44 status, the best spray window and free labour teams nearby. Each tile opens its full screen.

6. **One quick question a day**
   Optional, one tap, with the reason shown. Answers build village-level pictures (pollination, pests, crop size) that come back to growers as alerts.

SEASONS AND LANGUAGES

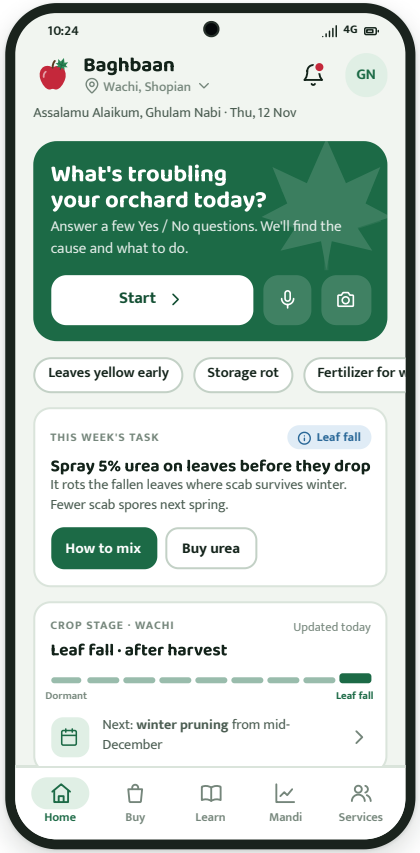
The same home screen in May, in November and in three languages

Content follows the orchard calendar and the language follows the grower. Urdu and Kashmiri modes switch the whole layout to right-to-left.



A May · petal fall

Scab risk, petal-fall spray, bees during bloom.



B November · leaf fall

Urea on leaves against next year's scab, frost alerts, CA storage space.



C Urdu

Right-to-left layout, Nastaliq type, Urdu numbers kept in Western digits for prices.

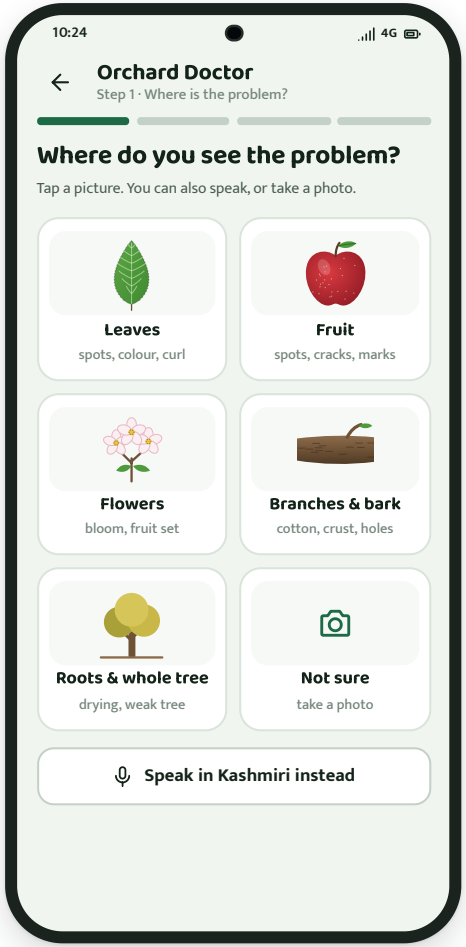


D Hindi

For non-Kashmiri workers, traders and family members who help.

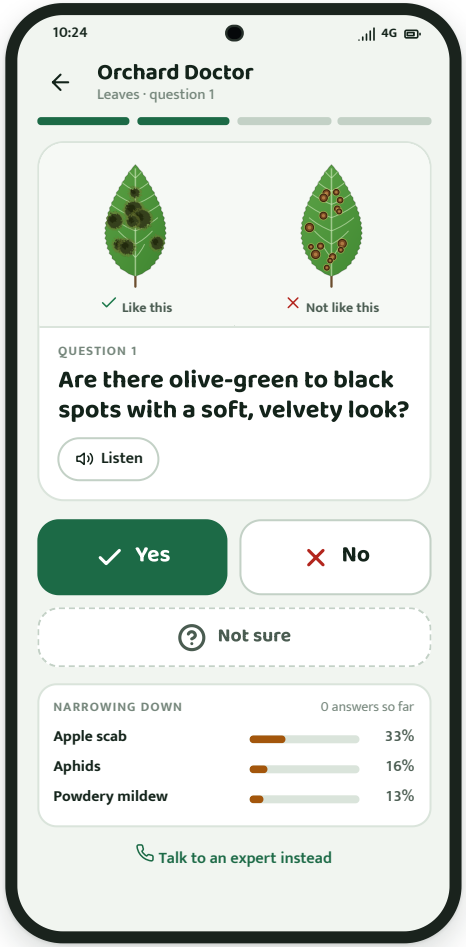
Yes or No, with a picture for every question

The grower never has to name a disease. They point at where the problem is and answer short questions. Each picture shows what a Yes looks like next to what a No looks like.



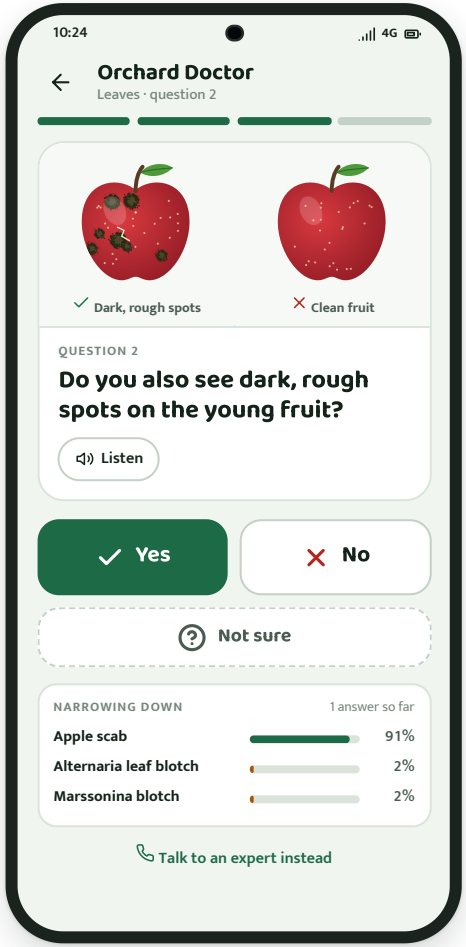
1 Where is the problem?

Leaves, fruit, flowers, bark or roots. Not sure? Take a photo or speak.



2 The best next question

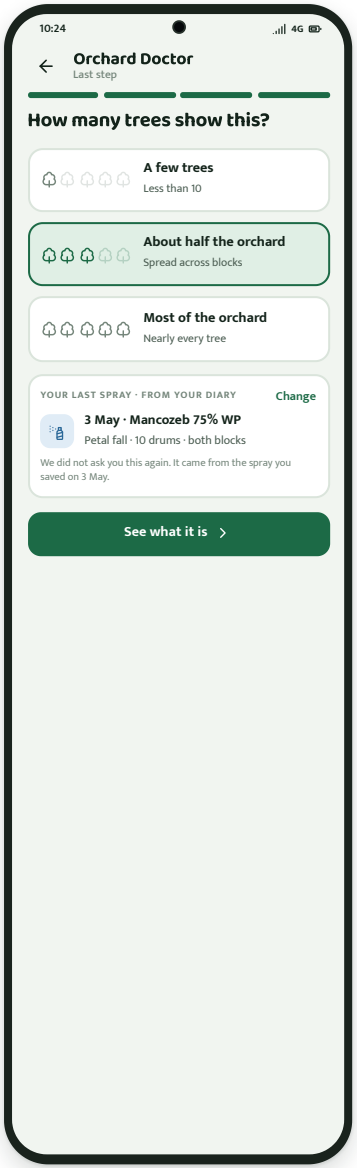
In May the likely causes on leaves are scab, aphids or mildew. The question that splits them best is asked first.

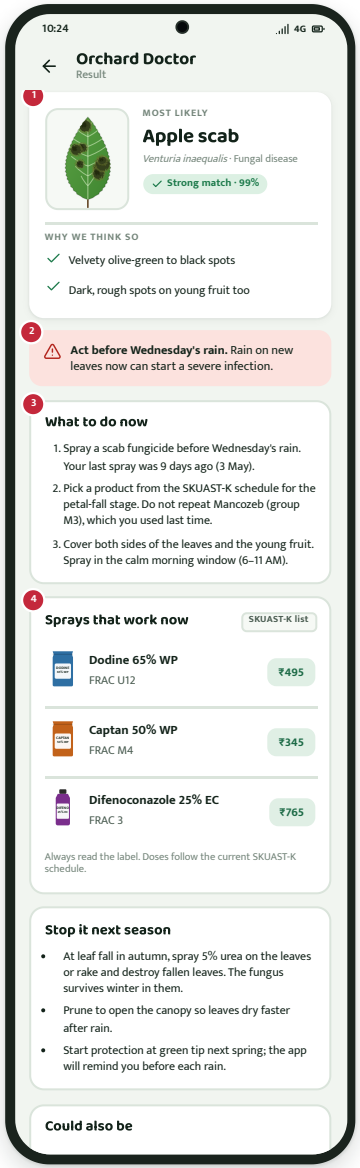


3 Narrowing down, live

One Yes moves scab from 33% to 91%. The next question checks the fruit to confirm.

A clear answer: what it is, what to do now, how to stop it





1. **Likely cause, with the reasons**
   The answers that led here are listed, so the grower can see why. Match strength is shown honestly.

2. **Urgency from the weather**
   The result knows tomorrow's rain. Acting before it is what matters most for scab.

3. **Steps in plain words**
   Personal: it knows the last spray from the diary and warns against repeating the same fungicide group.

4. **Products that fit**
   Only products with a label claim for apple and on the university schedule. One tap to buy from a licensed dealer.

5. **Expert and diary**
   Escalate with all answers attached, or save the case to the spray diary. "Could also be" covers look-alikes.

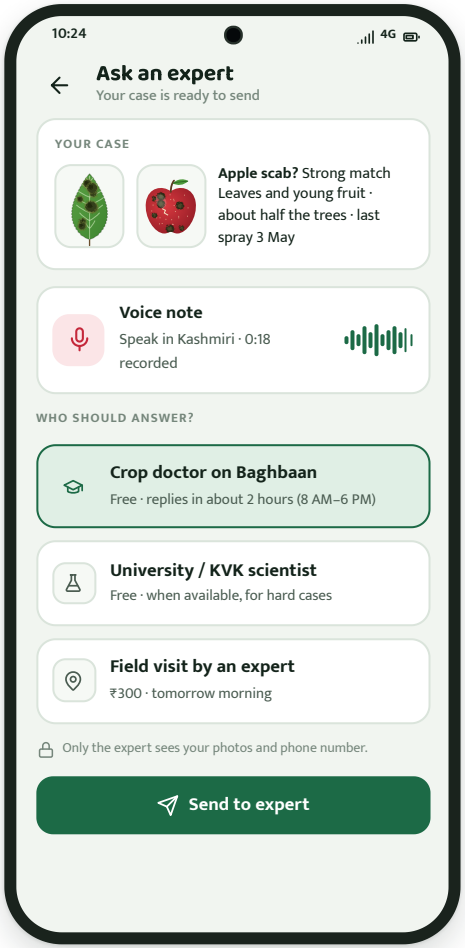
When answers are not enough: photo, expert, helpline

Questions work for most cases and need no data connection. Photos and people take over for the rest.



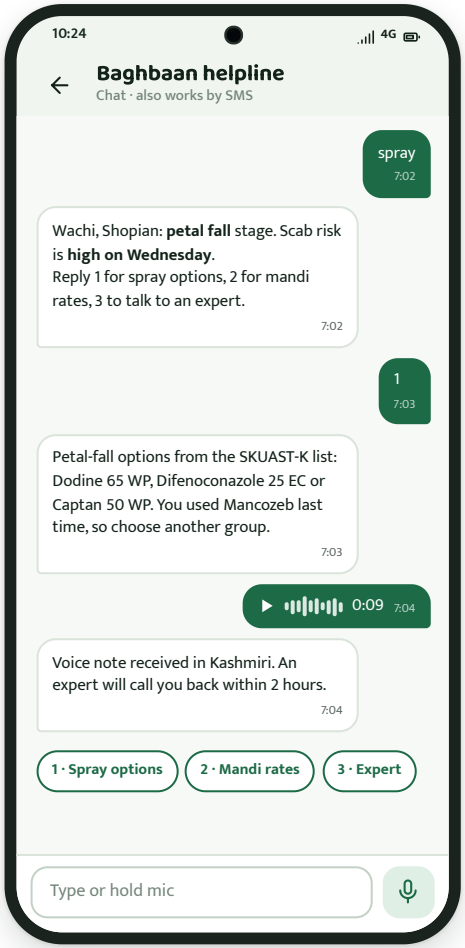
1 Photo check (beta)

An on-phone model trained on Kashmir leaf images. It confirms, it does not replace, the questions.



2 Ask an expert

The case, photos and a Kashmiri voice note go to a crop doctor. Hard cases go to university or KVK scientists.



3 Same help by SMS or chat

Keywords like SPRAY or RATE work on any phone. Voice notes are answered by call-back.

HOW THE DOCTOR THINKS

A transparent scoring engine, written with experts

The Doctor is a small, auditable model, not a black box. Experts write the questions and how likely each answer is for each problem. The app does the arithmetic.

1. **Start from what is common now**
   Each problem has a starting score for the season, altitude and recent reports nearby. In May, scab leads on leaves.

2. **Update with every answer**
   A Yes or No multiplies each problem's score by how often that symptom appears with it (naive Bayes). "Not sure" changes nothing.

3. **Ask the most useful next question**
   The app picks the question that splits the remaining problems best (largest expected information gain), so most cases end in 2–4 questions.

4. **Stop, or hand over**
   It stops when one cause passes 90%, and hands over to an expert when answers conflict or nothing fits.

5. **Learn from every case**
   Expert confirmations and photo labels tune the scores each season. Every version of the question bank is signed off by the content partner.

WORKED EXAMPLE FROM THIS DEMO · LEAVES, MID-MAY

POSSIBLE CAUSE	START	AFTER "VELVETY SPOTS: YES"	AFTER "SPOTS ON FRUIT: YES"
Apple scab	33%	91%	99%
Aphids	16%	1%	0%
Powdery mildew	13%	1%	0%
Caterpillars	9%	1%	0%
Red spider mite	8%	1%	0%
Alternaria leaf blotch	6%	2%	0%

Numbers are computed live by the prototype's engine with sample likelihoods. Real values will come from the content partner and labelled cases.

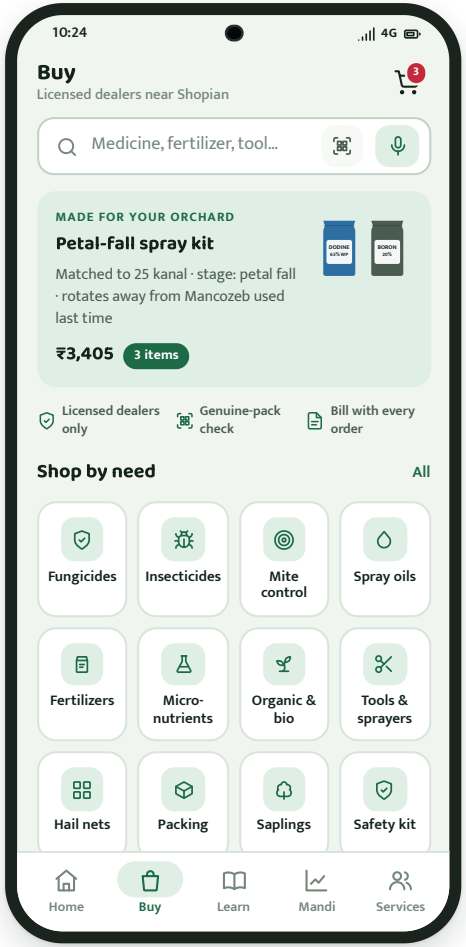
Safety rules in the engine

- Only products registered for apple and on the current university schedule can be suggested.
- Never an insecticide during bloom; never a banned or restricted molecule.
- Root rot, fire-blight-like symptoms or a sudden spread always go to a human.

BUY · 1

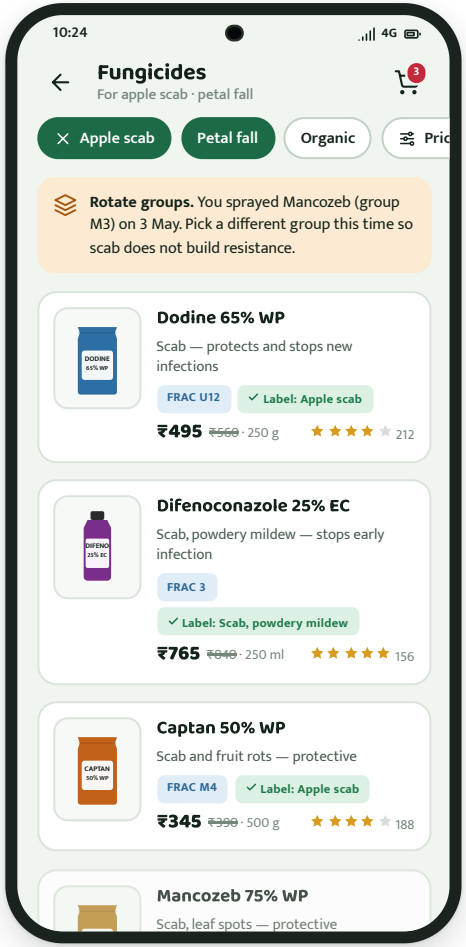
A shop that knows the orchard

The shop opens on a kit matched to the stage, the orchard size and the last spray. Products are sorted by problem, not by brand.



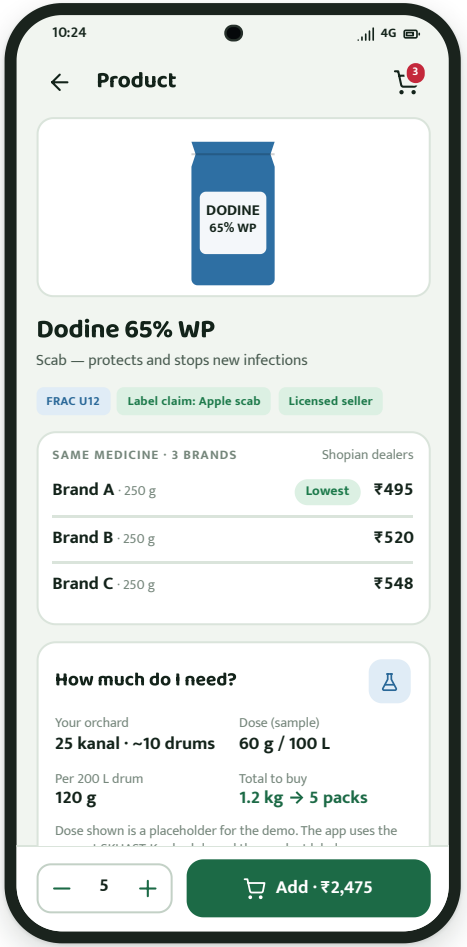
1 Made for your orchard

Kit for petal fall, sized for 25 kanal, rotated away from the last fungicide group.



2 Filtered by problem and stage

Each card shows the resistance group and the label claim, and greys out the group used last time.

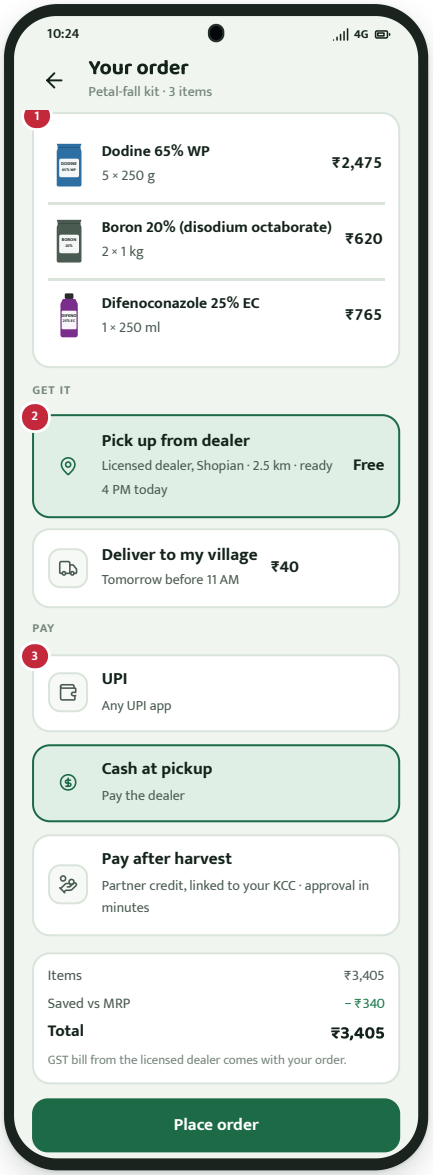


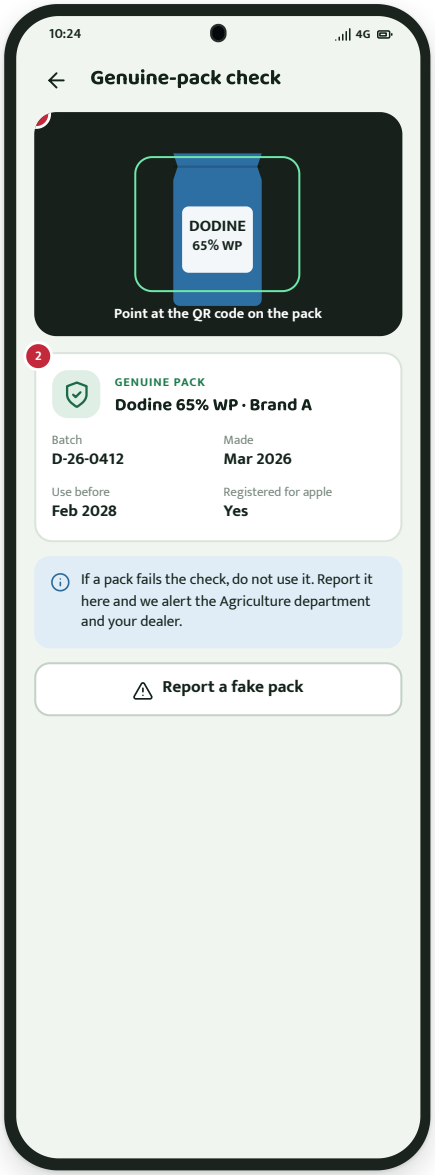
3 Compare brands of the same medicine

Price per pack across licensed dealers, the dose for the grower's drums, safety and verified reviews.

BUY · 2

Pay later, and know the pack is genuine





1 **Order from a licensed dealer**

Pesticides can only be sold by licensed sellers. Baghbaan is a marketplace for them; the dealer's GST bill comes with every order.

2 **Pickup or delivery**

Pickup at the village dealer keeps costs low and keeps dealers in the system instead of fighting it.

3 **Pay after harvest**

Partner credit linked to the Kisan Credit Card replaces informal dealer credit, at known interest, approved in minutes.

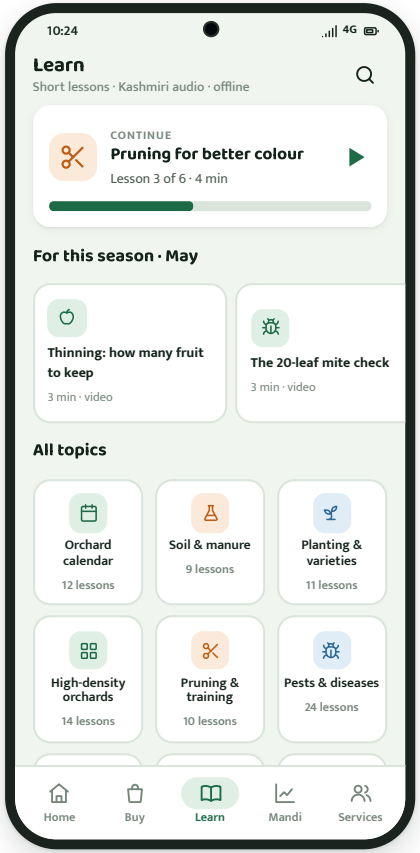
4 **Genuine-pack check**

Scan the QR on the pack: batch, dates and apple registration are checked against the maker's record. Fakes are reported to the department.

LEARN

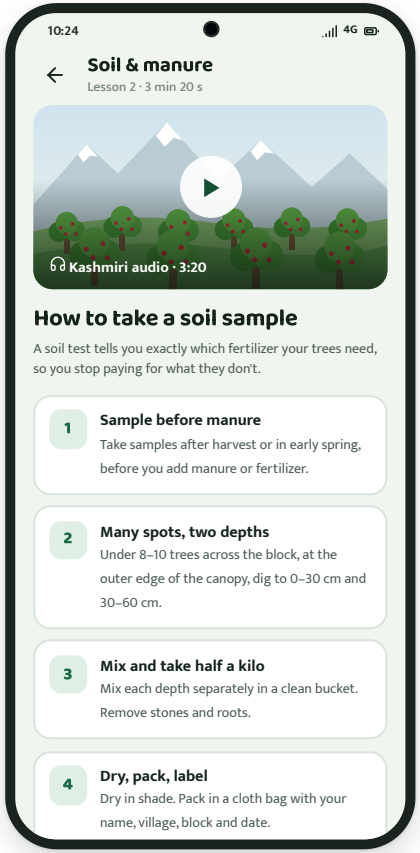
Short lessons in Kashmiri, saved on the phone

Three-minute videos and audio, step cards with pictures and a quick check at the end. Lessons appear when they matter in the season.



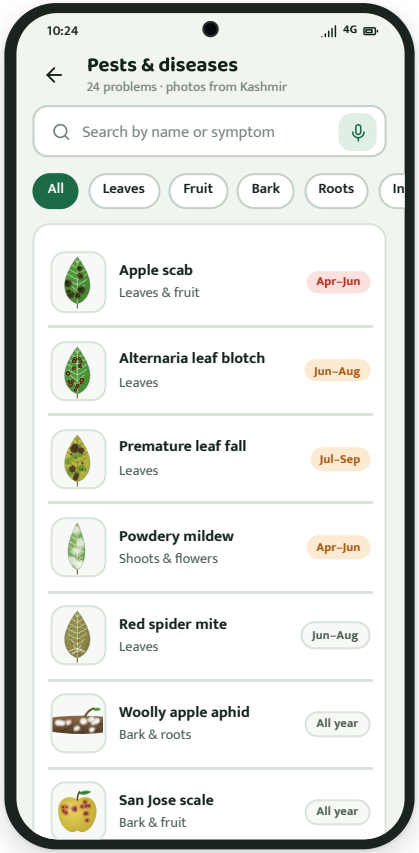
1 For this season

Thinning, the 20-leaf mite check, safe spraying near bees.



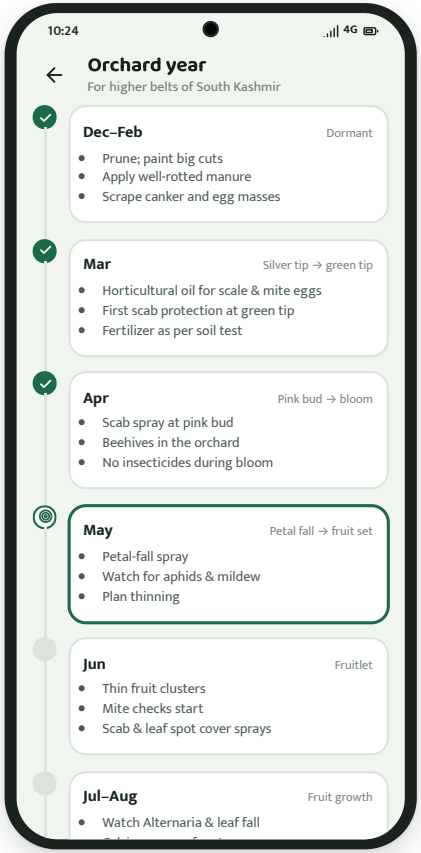
2 Step by step

Soil sampling in four picture steps, with one-tap booking of a soil test.



3 Pests & diseases

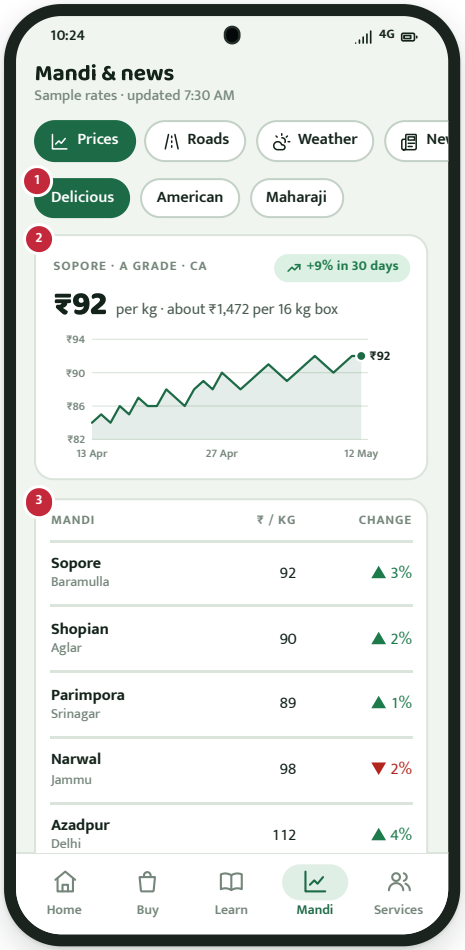
Every problem with its season, look-alikes and what to do.

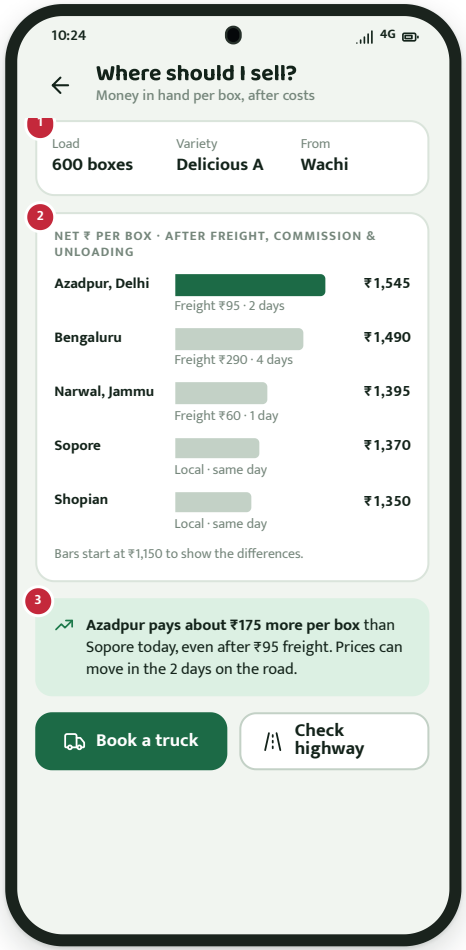


4 Orchard year

Month by month, adjusted for higher and lower belts.

Rates that end in a decision: where should I sell?





1. **Variety and grade**
   Rates per kg with the per-box figure next to it, because growers still think in boxes. Box weight is stated.

2. **30-day trend**
   Tap the chart for any day. The source and update time are always shown.

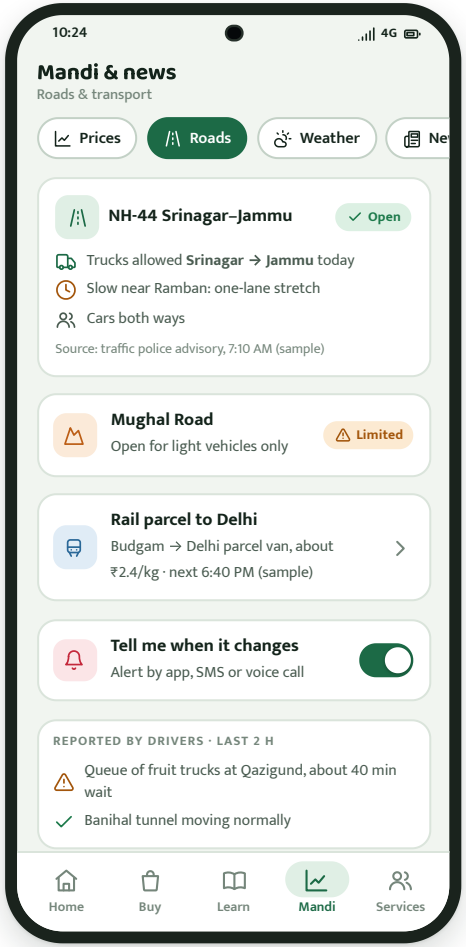
3. **All the mandis in one table**
   Sopore, Shopian, Parimpora, Narwal, Azadpur and other destination markets.

4. **Where should I sell?**
   Money in hand per box after freight, commission and unloading, for the grower's own load. Then one tap to book the truck.

ROADS, WEATHER AND NEWS

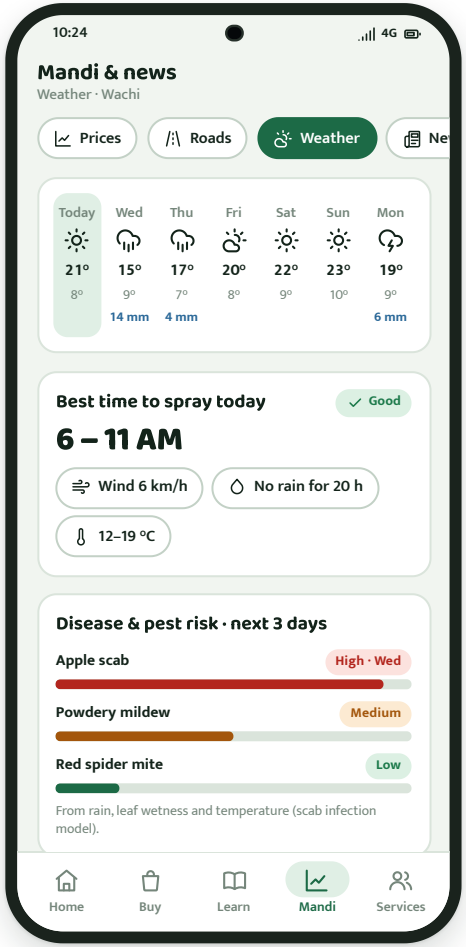
The things that decide a harvest week

The 2025 NH-44 closure stranded thousands of fruit trucks. Highway, rail and weather now sit next to prices.



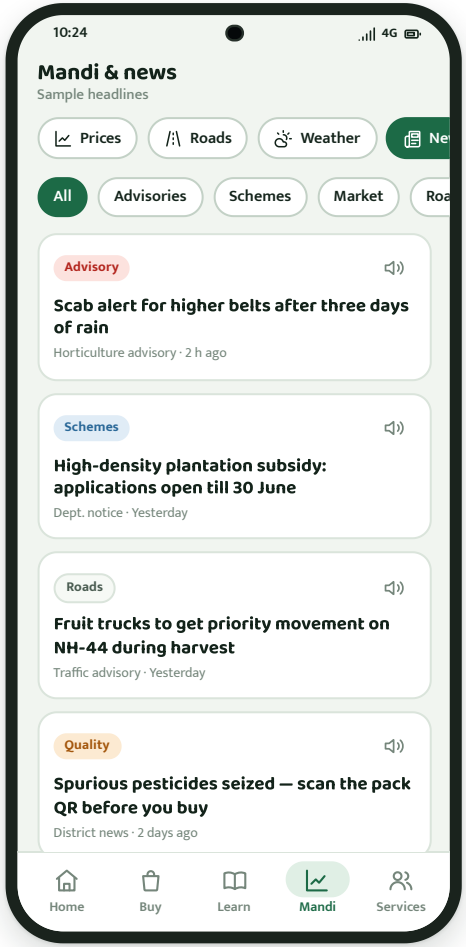
1 Highway, Mughal Road, rail

Official advisory plus drivers' reports. Alerts by app, SMS or voice call when status changes.



2 Spray window and disease risk

Best hours to spray today, and scab, mildew and mite risk from local weather.



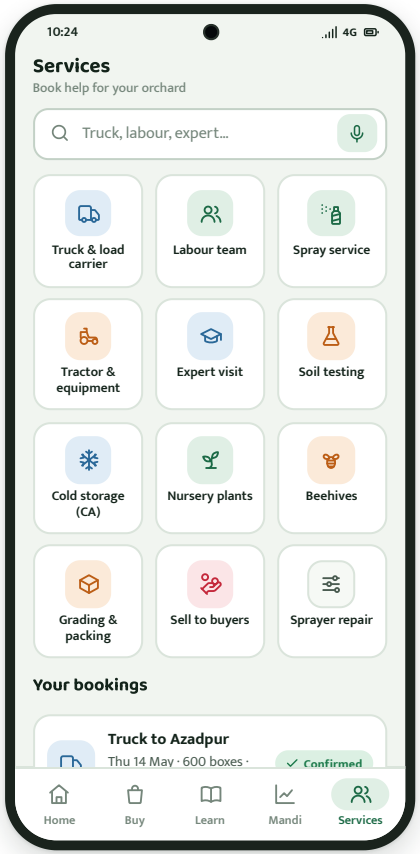
3 Advisories and schemes

Department advisories, scheme deadlines, quality alerts. Every item can be listened to.

SERVICES

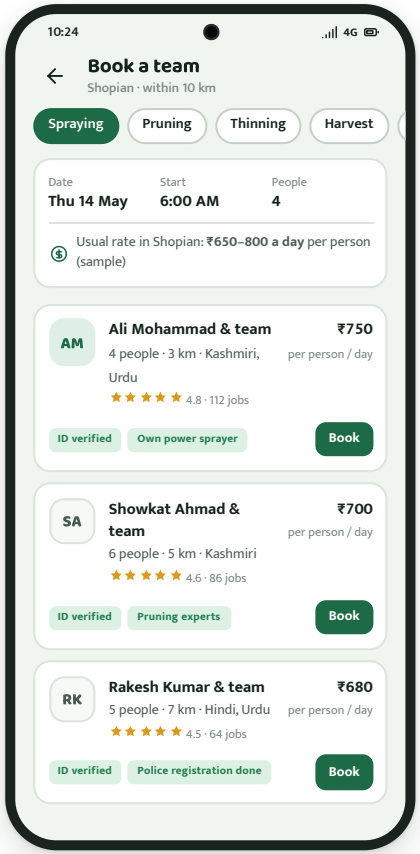
Book trucks, teams and experts, safely

Verified providers, fair-rate guides, shared loads and live tracking. Growers pay after the work is done.



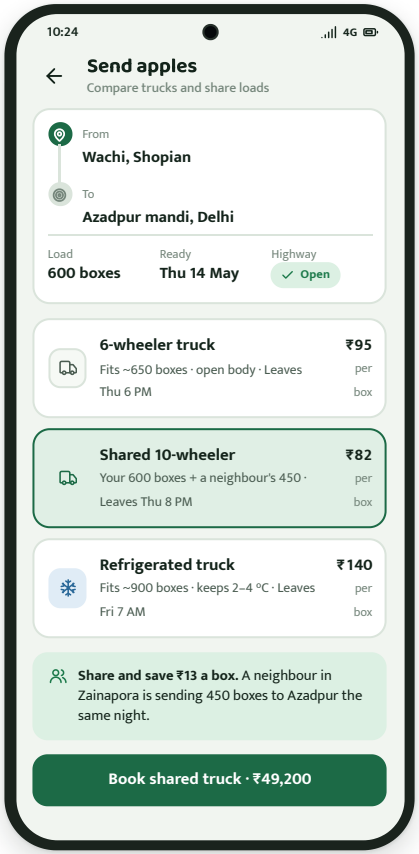
1 12 kinds of help

From trucks to beehives, cold storage and buyers.



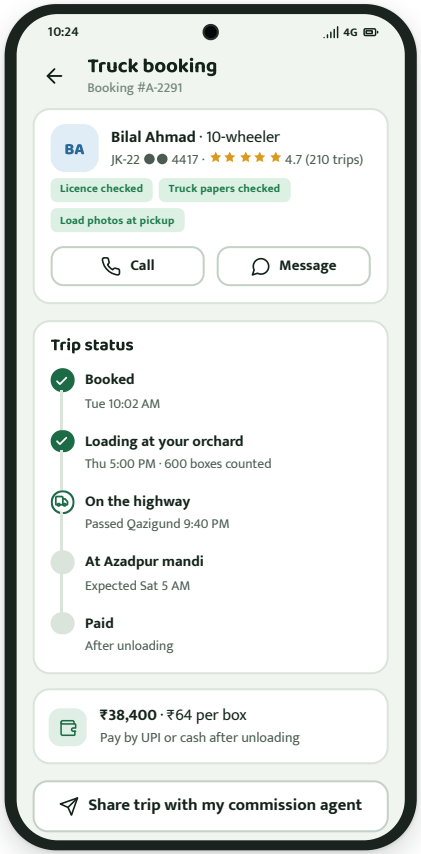
2 Labour teams

Rates, ratings, skills; non-local workers shown with police registration.



3 Share a truck

Pool part-loads with neighbours going to the same mandi.



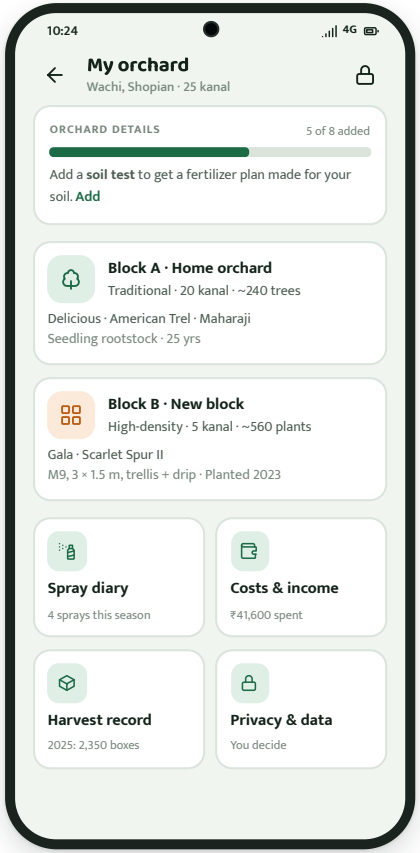
4 Track the trip

Checked driver and papers, live status, pay on delivery.

MY ORCHARD

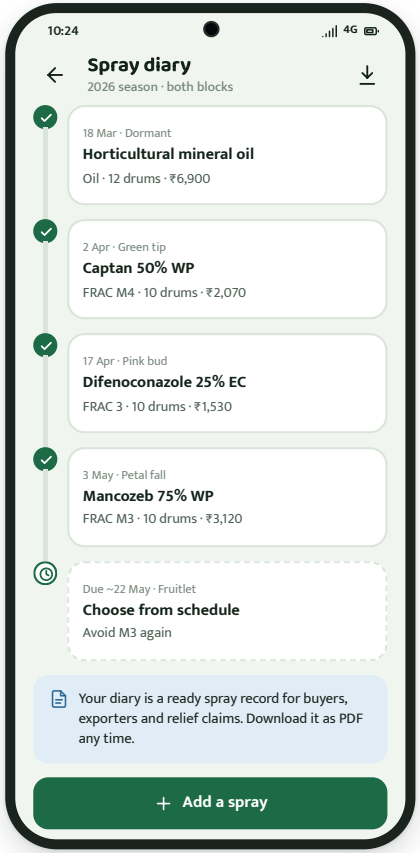
A record that builds itself

Blocks, sprays, costs and harvests, filled in from what the grower already does in the app. The record becomes useful to buyers, exporters and banks.



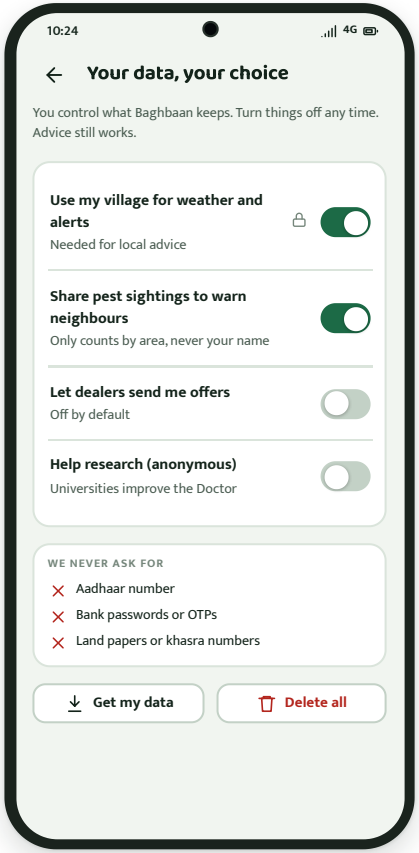
1 Orchard details

Optional, filled step by step. The meter shows what each detail improves.



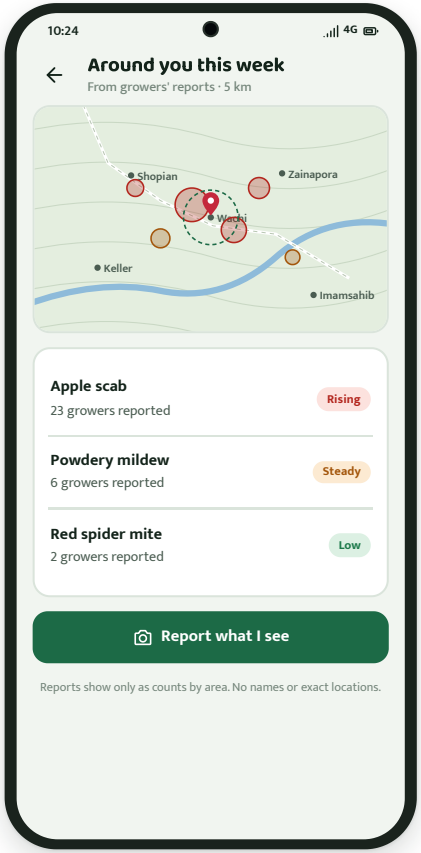
2 Spray diary

Every spray with group, drums and cost. Exports as a PDF record.



3 Your data, your choice

Plain switches, nothing shared with dealers by default, delete any time.



4 Pest map near me

Counts by area only. Sharing a report brings back an alert.

LISTENING WITHOUT PRYING

How the app learns about growers without a single form

Each question is asked once, at the moment it improves an answer, with the reason shown and a Skip. What the app already knows is never asked again.

WHEN	WHAT WE ASK	WHAT THE GROWER GETS BACK
First open	Language, village	Local weather, crop stage, rates
First diagnosis	Where the problem is, how many trees	A cause and a plan; urgency
First purchase	Orchard size (kanal), sprayer drum	Exact quantity to buy; less waste
Each spray	Product, date (tap from the order)	Group rotation, next-due reminder, spray record
Once a day, optional	One tap: bees seen? pests seen?	Village alerts built from everyone's taps
End of season	Crop better, same or worse?	Area crop estimate, price outlook
Only if they want	Phone number, name	Expert call-back, bookings, credit

1. **Value before questions**
   Every screen is useful in guest mode. Questions come after the app has helped.

2. **Never asked**
   Aadhaar, bank details, OTPs or land papers. Payments run through licensed partners.

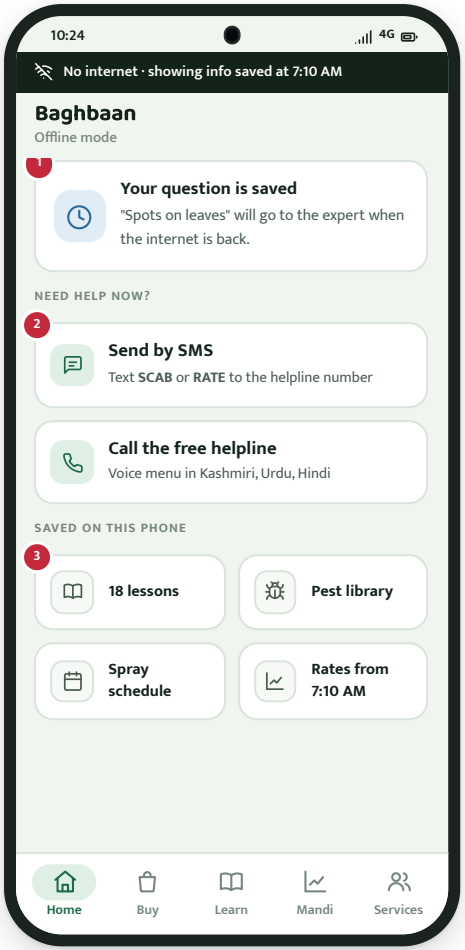
3. **Aggregates only**
   Pest and crop reports leave the phone as counts by area, never as names or exact locations.

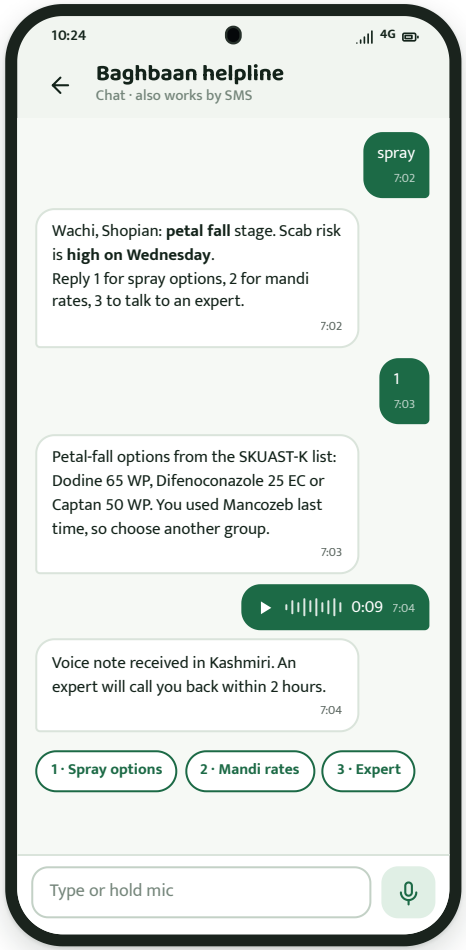
4. **Ready for India's data law**
   Itemised consent notice in Kashmiri, Urdu, Hindi and English; withdrawal as easy as consent; parental consent under 18; export and delete. The DPDP Rules' main duties apply from May 2027.

5. **Shared phones, family helpers**
   Many households share one phone. A "helping someone" mode keeps the grower's own consent.

LOW CONNECTIVITY

Works when the internet does not





1. **Questions wait, they do not fail**
   A diagnosis runs fully on the phone. Expert questions and orders queue and send when a signal returns.

2. **SMS and a voice line**
   Keyword SMS (SPRAY, RATE, ROAD) and a missed-call voice menu in Kashmiri, Urdu and Hindi reach basic phones too.

3. **Saved on the phone**
   Lessons, the pest library, the spray schedule and the last rates are cached and dated.

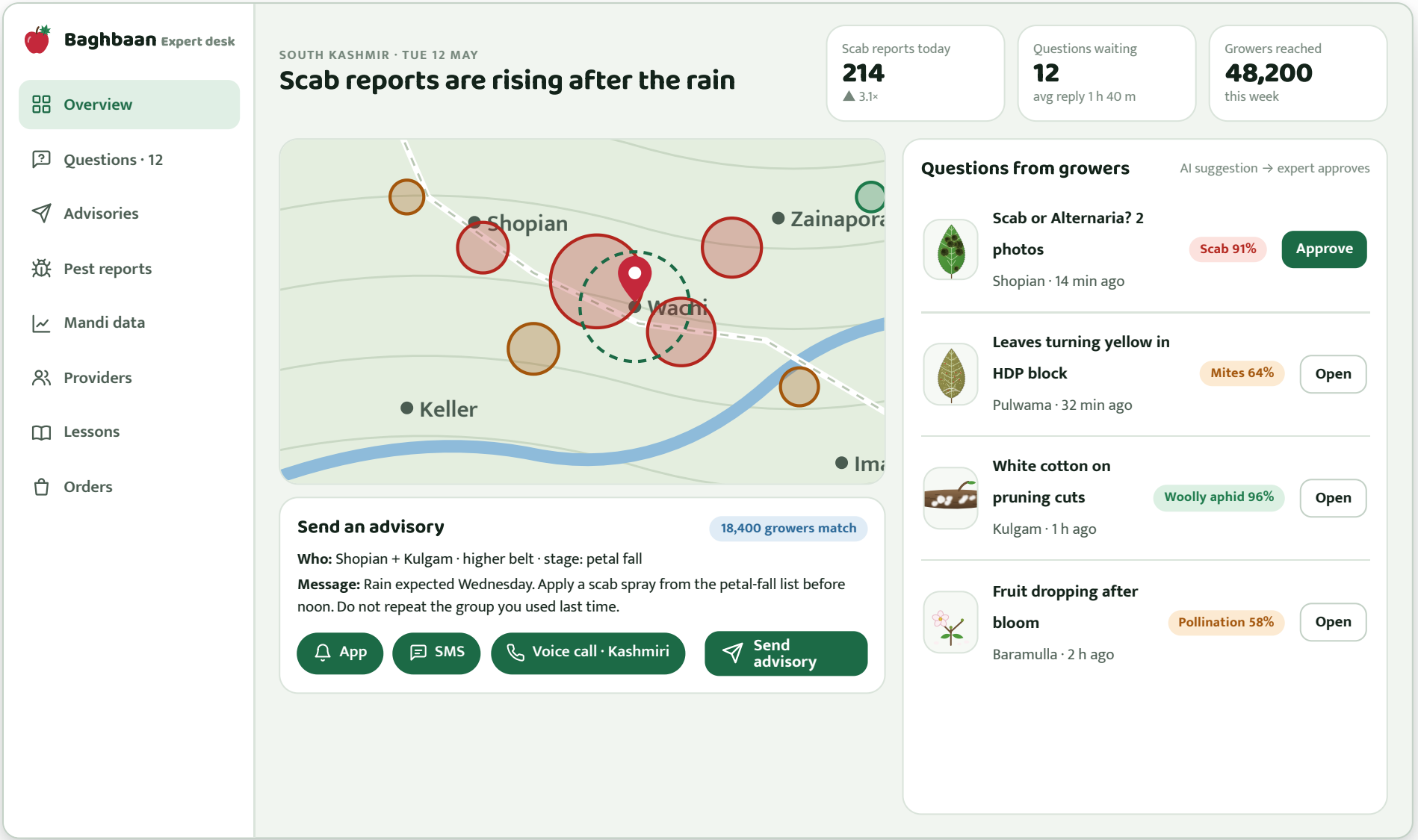
Why it matters here

Jammu & Kashmir has had more internet shutdowns than any other part of India, including a 2019–2021 period of 550 days with no or partial mobile internet. An orchard app that only works online fails exactly when growers need it.

BEHIND THE APP

The expert desk: one place to answer growers and warn them early

Crop doctors approve AI-suggested answers, watch reports rise on the map and send advisories by app, SMS and Kashmiri voice call to exactly the growers who need them.



ROADMAP

From a pilot in two districts to the whole valley

Build what earns trust first (diagnosis, rates, roads, lessons), then the marketplace, then credit and data services.

PHASE 1 · MONTHS 0–6

Pilot: Shopian and Baramulla

- Orchard Doctor with 25 problems, question bank reviewed by a university partner
- Mandi rates (Agmarknet / data.gov.in), highway and weather alerts
- 40 lessons with Kashmiri audio; offline mode; SMS keywords
- Directory of verified transporters and labour leaders; booking by call
- 500 growers recruited through village helpers and Kisan Khidmat Ghars

PHASE 2 · MONTHS 6–12

Marketplace and records

- Licensed-dealer shop with pickup, delivery and genuine-pack checks
- Spray diary, group rotation and dose calculator
- In-app truck, team and cold-storage booking with tracking
- Photo check (beta) trained on Kashmir datasets
- Voice helpline in Kashmiri; expert desk for partners

PHASE 3 · MONTHS 12–24

Scale and services

- All apple districts; Ladakh and Himachal content later
- Pay-after-harvest credit with a bank or NBFC; insurance if launched
- Weather-station network for disease forecasting
- Buyer and exporter links using spray records
- Anonymised pest and crop intelligence for the department

Partners to approach

Content: SKUAST-Kashmir, ICAR-CITH Srinagar, KVKs. Advisories and schemes: Department of Horticulture. Supply: licensed dealers and manufacturers. Logistics: transporter unions, rail parcel. Money: banks/NBFCs for KCC-linked credit. Reach: FPOs, mandi associations, Kisan Khidmat Ghars.

What we will measure

Weekly active growers; diagnoses completed and confirmed right by experts; sprays saved by avoiding wrong products; share of genuine-checked purchases; truck days lost to closures avoided; repeat bookings; grower-reported net price per box.

ABOUT THIS DEMO

What is working, what is sample, what needs checking

Working in the clickable demo

- All 34 screens and navigation between them
- The Orchard Doctor engine: real scoring and question choice for leaves, fruit, flowers, bark and roots
- Language switch with right-to-left Urdu and Kashmiri modes
- Chart hover, toasts, bookings and booking states

Sample data

- All names, providers, prices, mandi rates, freight, headlines and reviews
- Weather, disease risk and highway status
- Likelihood values in the question bank
- Product brands (shown as Brand A, B, C)

To verify before launch

- Every dose and product choice against the current SKUAST-K spray schedule and product labels
- Kashmiri voice recordings by native speakers
- Legal review: pesticide sales licensing, DPDP consent, platform-worker rules
- The name "Baghbaan" (trademark search)

Companion documents

The research and integration plan covers the market, the data sources and APIs, the technologies, datasets, regulations, competitors and partners in detail. Clickable demo: **claude.ai/artifact/T7wV1p9vo63Sqa659bEQCG**. Source code: **prototype/** in the project repository.



Bagh ka har kaam, ek app mein.

Every orchard job, in one app. Next step: put this demo in front of twenty growers in Shopian and Sopore, and let them tell us what to change.